

Reading a phase diagram before calculating

Read and prepare: 15–20 minutes total. Bring tentative answers; uncertainty is useful for discussion.

At fixed temperature, how does lowering pressure change a binary mixture?

The state and the constraints

Start by naming the components, phases and imposed conditions. For a TP flash, temperature T , pressure P and overall mole fraction z are given. The calculation finds the phase amounts and compositions. The liquid composition x , vapor composition y and overall composition z have different meanings. A bubble calculation supplies liquid composition; a dew calculation supplies vapor composition.

Reading P-x-y

At fixed T , the bubble curve bounds the liquid region and the dew curve bounds the vapor region. For the ideal example below, pressure above the bubble curve gives liquid and pressure below the dew curve gives vapor. A horizontal line through the two-phase region connects coexisting x and y . A vertical line at z follows changes in pressure at fixed overall composition. The position of z along a tie line determines phase amounts.

Equilibrium and amounts

Equal component fugacities establish phase equilibrium. A material balance supplies the amount information. For a nonreacting binary with two phases, the phase rule gives two intensive degrees of freedom; specifying T and P fixes the coexistence compositions in an ordinary two-phase region. Overall composition then determines amounts. Degenerate states, such as an azeotrope, need special care.

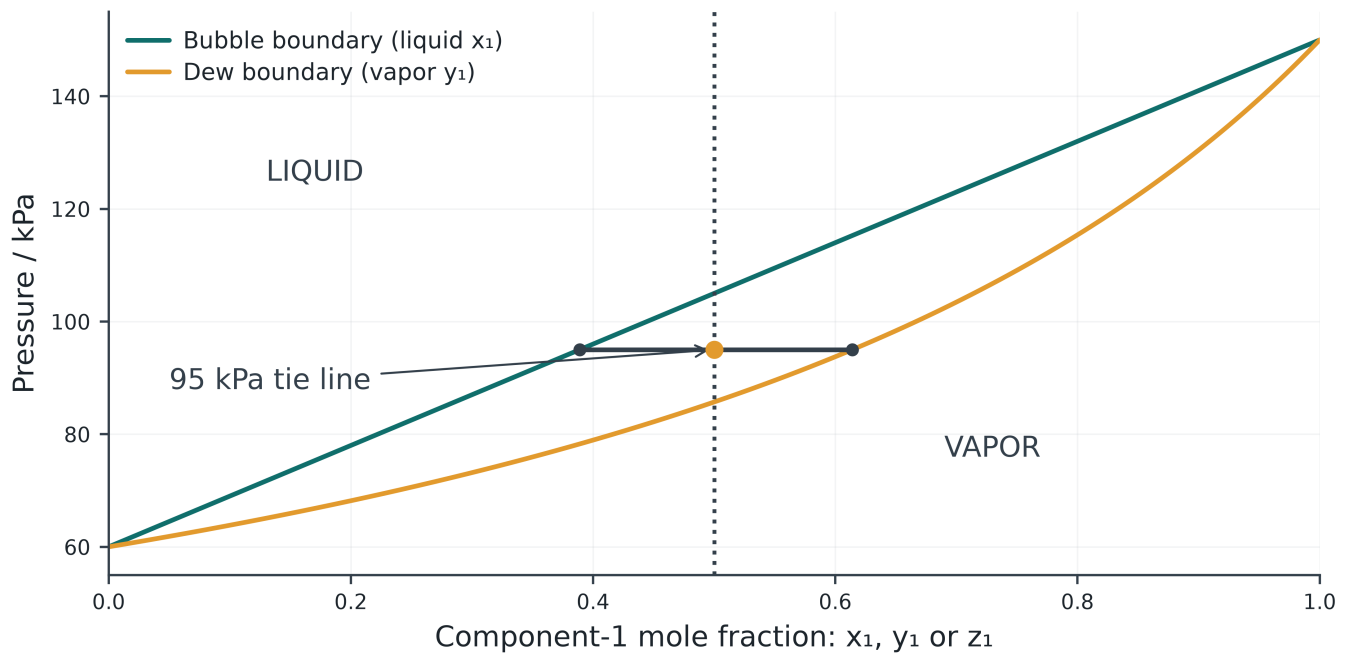
$$P_{\text{bubble}} = \sum x_i P_i^{\text{sat}}; 1/P_{\text{dew}} = \sum y_i / P_i^{\text{sat}} \text{ (ideal phases, fixed } T)$$

$$z_i = (1 - \beta)x_i + \beta y_i; \beta = (z_1 - x_1)/(y_1 - x_1)$$

One example to reason through

Synthetic teaching values. These numbers illustrate the assumptions; they are not measurements of a real material.

Use synthetic saturation pressures 150 and 60 kPa at 350 K, with $z_1=0.5$ and $P=95$ kPa. The bubble and dew pressures evaluated at composition 0.5 are 105 and 85.7143 kPa. Thus 95 kPa lies inside the two-phase interval. The horizontal tie line gives $x_1=0.38889$ and $y_1=0.614035$. The lever rule gives $\beta=0.493506$. Substitution closes the component-1 balance to rounding. The second component must also balance.



Check and interpret

At 110 kPa the same feed is liquid in this model. At 80 kPa it is vapor. Do not apply the two-phase lever rule outside the coexistence interval.

Bring three short answers

A sketch, a few numbers or one sentence is enough. Do this before opening the calculator; revise your prediction in class.

1. Sketch where the bubble curve, dew curve and $z=0.5$ lie. What does a horizontal line tell you?

2. Why can the same numerical value 0.5 mean different inputs in bubble, dew and flash calculations?

3. Predict the phase at 110 kPa before opening Lab 01. Which quantities remain specified?

Optional extension

After the ideal case, compare modified Raoult predictions using a liquid activity model. Virial vapor corrections and an adiabatic energy balance are optional extensions in Labs 09 and 07. Cubic-EOS VLE is reference material, not an implemented lab solver.

Continue in the core lab

<https://www.skhgroup.net/teaching/2105603/labs/>

Reading locator: supplied textbook contents, sections 15.1-15.4, 15.9-15.12.

Original course explanation and examples, not a reproduction of the textbook. Section numbers are based on the supplied contents pages. Required weekly independent work: 15–20 minutes preparation plus 30–40 minutes follow-up; optional extensions are additional.